TEXORA · LECTURE SHEET

Dyeing Processes, Garment Dyeing and Faults

A dyeing process is Topic 1's physical chemistry scheduled in time: exhaustion, fixation, washing-off, each with its temperature and dosing curve. This sheet sets out the standard processes, the special case of dyeing made-up garments, and a diagnostic table of the faults that result when the schedule is broken.

WP-4101 · TEXTILE CHEMISTRY & COLORATION-III · TOPIC 3 · SPECIALIZATION — WET PROCESSING

1. Anatomy of an exhaust dyeing process

Nearly every batch process follows the same five acts, whatever the dye class:

1. **Preparation in the bath:** goods wetted out with auxiliaries; dye dissolved/dispersed and added in a form that cannot spot the goods.
2. **Exhaustion:** temperature raised at a controlled rate (or held) while dye transfers to the fibre; salt dosed to drive exhaustion for anionic-on-cellulose systems.
3. **Fixation:** the step that makes the colour permanent — alkali for reactive dyes, oxidation for sulphur, cooling/thermosol history for disperse.
4. **Washing-off:** removal of unfixed and surface dye — rinses, then hot soaping; for reactive dyes this stage decides wash fastness.
5. **Aftertreatment:** cationic fixers, softeners, or fastness improvers as specified.

The craft lies in the dosing and temperature curves inside acts 2 and 3: alkali dumped at once fixes dye at the surface it first touches (unlevel, poor penetration); a linear or progressive dosing curve keeps the fixation rate matched to circulation (WP-3207 Topic 7).

2. The standard processes by class

CLASS	PROCESS OUTLINE (TYPICAL VALUES; PLANT RECIPES RULE)

Reactive on cotton (exhaust)	Enter dye and auxiliaries at 40–60 °C; exhaust with salt (30–80 g/L by depth) over 20–30 min; fix by dosing soda ash (5–20 g/L) progressively over 30–45 min at the dye's temperature class; hold 30–60 min; drain, rinse, hot soap at the boil, rinse. Pad routes: pad–dry–cure, pad–steam, cold pad–batch.
Disperse on polyester (HTHP)	Acidic bath (pH ~4.5–5.5), dispersing agent; raise 1–2 °C/min to 130 °C; hold 30–60 min; cool slowly to avoid oligomer and crease faults; reduction-clear for deep shades to remove surface dye; rinse.
Sulphur on cotton	Reduce dye with sodium sulfide in hot alkaline bath; dye at 90–95 °C with salt for exhaustion; rinse; oxidise (peroxide or bromate systems) in a fresh bath; soap, rinse. Modern pre-reduced liquid sulphur dyes and low-sulfide reducing systems follow the same outline with cleaner chemistry.
Azoic on cotton	Naphtholate: impregnate with naphthol solution (+ alkali, wetting agent, salt), remove excess liquor uniformly; develop in cold diazo salt bath (buffered near-neutral); rinse, hot soap to remove surface pigment and develop true shade and rub fastness.
Pigment (pad)	Pad pigment + binder + auxiliaries at controlled pick-up (WP-3207 Topic 4 arithmetic); dry carefully to prevent migration; cure 150–160 °C for the binder's specified time.

3. Bulk correction — the addition calculation

WORKED EXAMPLE 1 — SHADING ADDITION ON A BULK LOT

A 600 kg reactive-dyed lot measures slightly weak in the red component; colour-physics correction calls for an addition of 0.12% owf of the red dye.

Addition = 0.12 × 600/100 = **0.72 kg** of red dye, added as a prepared solution through the dosing system, with a shortened re-fixation (alkali already largely consumed: the colourist checks bath pH and tops up alkali only per the plant's correction procedure).

Corrections are made on measured ΔL/Δa/Δb (Topic 3, WP-3207), never by eye at the machine — and each addition is recorded against the lot so the next bulk of the same shade starts closer (right-first-time learning).

4. Garment dyeing

Dyeing the **made-up garment** inverts the usual order: fabric is prepared, cut and sewn in an undyed (PFD — prepared-for-dyeing) state, and colour is applied last. Consequences and requirements:

- **Why:** colour can be decided late — near the selling season — and unsold colours are not stranded as dyed fabric; garment-dyed aesthetics (soft wash-down, seam puckering contrast) are themselves a

style.

- **Preparation:** garments must be uniformly desized/scoured (often in the dyeing machine itself); sewing thread must dye with the garment (cotton thread for cotton garments) or is deliberately chosen as contrast; labels, interlinings and elastane must survive the process.
- **Machines:** rotating perforated drums and paddle machines (WP-3207 Topic 10 machines serve here) with gentle action, higher LR than piece dyeing, and loads balanced so garments circulate freely; shrinkage must be engineered into the pattern, because the garment shrinks in its first dyeing.
- **Classes used:** reactive and direct for cellulosic garments; pigment garment dyeing for vintage effects (pigment lodges on surfaces and seams, then washes down); disperse routes are impractical for most made garments with mixed components.
- **Faults specific to garments:** seam and fold shading differences, abrasion marks at edges, component mismatch (button, zip tape, thread dyeing differently), and load-to-load variation demanding stricter batching by garment size and fabric lot.

5. Faults — diagnosis table

FAULT	PROBABLE CAUSE	REMEDY / PREVENTION
Unlevelness (patchy)	Too-rapid strike: additions dumped, temperature ramp too fast in the critical zone, poor circulation or overloaded machine	Progressive dosing, controlled ramp (Topic 1 kinetics), correct loading; strip and re-dye if severe
Ending / listing (jigger)	Bath exhausts between passages; temperature/concentration gradient across the trough	More passages, top-up dosing, pre-dissolved additions; check cloth temperature uniformity
Barré (stripes in knits)	Yarn-lot differences in the fabric itself — dye uptake varies between feed yarns	Not a dyeing fault: segregate yarn lots; dye selection with good covering power; inform fabric sourcing
Crease marks	Goods cooled or held folded under load; rapid cooling of thermoplastics	Controlled cooling, keep goods moving, correct loading density
Poor wash fastness (reactive)	Hydrolysed dye not soaped off; under-fixation from low alkali/temperature	Full soaping programme; verify fixation conditions; cationic fixer only as specified supplement, not as rescue
Shade change on drying / store	Metamerism (WP-3207 Topic 3); thermomigration of disperse dye in	Check under all illuminants; control heat-set temperature; correct sulphur oxidation and aftertreatment

	heat setting; sulphur tendering/bronziness	
Specks and spots	Undissolved dye, hard-water precipitates, silicone spots from auxiliaries	Proper dye preparation and straining; sequestering agent; auxiliary compatibility checks
Two-sided shade (face/back)	Pad pick-up difference between faces; migration during drying	Balance nip pressures; controlled intermediate drying; anti-migrant in pad liquor

6. Glossary — key terms

- **PFD:** prepared for dyeing — garments made from undyed, prepared fabric for later garment dyeing.
- **Reduction clearing:** alkaline reducing bath removing surface disperse dye after HTHP dyeing.
- **Soaping:** hot detergent treatment removing unfixed dye after dyeing.
- **Thermomigration:** movement of disperse dye within/on polyester at high dry temperatures, altering shade and fastness.
- **Bronziness:** metallic surface sheen on heavy sulphur dyeings from surface dye.

7. Key points

- Every process is exhaustion → fixation → wash-off scheduled in time; faults are usually schedule failures, not dye failures.
- Garment dyeing moves colour to the end of production — commercially agile, technically demanding on components and shrinkage.
- Diagnose faults from mechanism: the table's cause column is Topic 1 and Topic 2 read backwards.

8. Summary

Process competence is the ability to keep hundreds of kilograms of goods on the scheduled path through exhaustion and fixation, to correct a measured shade error with a calculated 0.72 kg addition, and to read a fault back to the act of the process that produced it. That, more than any single recipe, is what a dyehouse is paid for.

9. Practice

1. A correction requires +0.08% owf of a blue dye on a 450 kg lot, and +0.03% owf of a yellow on the same lot. Calculate both additions.

2. A reactive dyeing washes down badly in the customer's first wash although the shade was correct at the machine. Which act of the process failed, and what is the check?
3. Why is cotton sewing thread specified for garment-dyed cotton garments, and what happens if polyester thread is used?
4. Distinguish barré from unlevelness by origin, and state why no change of dyeing conditions can fully cure barré.

Answer key

1. Blue: 0.08 × 450/100 = 0.36 kg. Yellow: 0.03 × 450/100 = 0.135 kg (135 g).
2. Washing-off (act 4): hydrolysed, unfixed reactive dye was left on the goods and is now desorbing into the customer's wash. Check: extend/verify the hot soaping stage (time, temperature, detergent) and test wash fastness at the laboratory before release.
3. The thread must dye with the same dye class as the garment so seams match. Polyester thread does not take reactive/direct dyes: it stays white or takes only a stain, leaving undyed seams — a defect unless deliberately designed as contrast.
4. Barré originates in the fabric — yarns of different dye affinity knitted together — whereas unlevelness originates in the dyeing process (strike, circulation, dosing). Because the affinity difference is built into the goods, process changes can only mask it slightly (dye choice, longer levelling); the cure is yarn-lot control before knitting.

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